Evoke: A Framework for Measuring Emotional Arousal to Robotic Voice Modulation Using Electrodermal Activity

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Abstract: Many neurodivergent users react strongly to sound, and a robot's voice can become calming or overwhelming depending on how it is delivered. Designers need a clear way to measure how vocal pitch affects emotional comfort. This paper presents *Evoke*, an exploratory feasibility pilot that examines how robotic voice modulation influences emotional arousal. A Pepper robot read short story passages using three pitch settings. Electrodermal activity was recorded and paired with brief self-report surveys. Four participants took part in this study. Early patterns appeared in the data. Low-pitch narration produced the largest calming response, shown by reductions in tonic skin conductance. Default pitch also reduced arousal, although the results varied across individuals. High-pitched narration showed mixed effects and clear differences in how people interpreted the voice. These preliminary findings show that the protocol can capture changes in autonomic state during robot-delivered speech. *Evoke* provides an initial foundation for larger studies and may guide the development of comfortable robot voices for users with auditory sensitivity.

1 INTRODUCTION

For neurodivergent users, particularly those with attention-deficit/hyperactivity disorder (ADHD), how a robot speaks can influence more than conversational clarity alone. It can shape emotional comfort, focus, and trust during interaction. Socially assistive robots (SARs) are being recognized as valuable tools for supporting these populations, offering structured therapeutic and social engagement. Recent trials with humanoid robots such as *Pepper* show promising results for supporting users with ADHD (O'Connell et al., 2024). These robots provide a consistent and non-judgmental presence that many users found helpful. This kind of steady companionship is similar to what individuals experience during parallel task completion, where simply having another entity nearby can support focus and motivation (Park and Catrambone, 2012; Zajonc, 1965).

At the same time, it is well-established that individuals with ADHD often experience atypical sensory processing. Auditory sensitivity, in particular, is a common challenge for this group (Bijlenga et al., 2017). This leads to a central design question: how a robot communicates through tone, pitch, and vocal prosody may not be a one-size-fits-all solution. These vocal elements can significantly impact the comfort and emotional experience of users with ADHD. This suggests that robotic voice modulation must be examined with greater precision, especially when designing inclusive systems for neurodivergent users.

In the field of human–robot interaction (HRI), vocal prosody is widely known to influence both emotional response and user engagement. Grounded theories such as the Computers Are Social Actors (CASA) paradigm show that people apply human-like social expectations to machines (Nass and Brave, 2005a). Subtle shifts in pitch or vocal affect can shape how a user perceives a robot's intent, emotion, and personality. Previous work has shown that higher-pitched or more animated speech tends to boost engagement. In contrast, lower-pitched tones are often interpreted as calm or serious (Rodero and Blanco, 2022). These effects are consistent with well-documented human communication patterns. However, deeper analysis is

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still needed for user groups that may be particularly susceptible to these shifts in vocal delivery, including individuals with ADHD.

As part of preliminary work, we developed and tested a structured evaluation protocol to evaluate this idea. A *Pepper* robot delivered identical narrative content under three distinct voice conditions: high, low, and default pitch. Each session was designed to measure electrodermal activity (EDA) as an indicator of emotional arousal, using an Empatica E4 wristband. Although the sample size was small and participants were not specifically diagnosed with ADHD, the data revealed consistent physiological changes in response to vocal pitch. This early signal suggests that robotic voice modulation has a measurable emotional effect that warrants further exploration, especially within the ADHD population.

Those with ADHD often experience heightened emotional responses to sensory stimuli (Bijlenga et al., 2017). These reactions can impair attention, increase stress, or disrupt task completion. This makes voice modulation in SARs more than just a stylistic choice because it becomes a design constraint. We argue that vocal delivery should be guided by objective measures such as EDA. Unlike self-report methods, EDA provides a continuous, real-time view of physiological arousal (Boucsein, 2012). Through this lens, researchers can assess how specific voice profiles affect user state over time. This opens the door to adaptive robot speech patterns that modulate in response to user engagement or overstimulation.

To address this gap, we explored a method for linking robotic voice modulation with emotional arousal using electrodermal activity (EDA). Our goal was not to build a final model or make broad claims. Instead, we aimed to explore the feasibility of a reproducible and structured protocol that observes how variations in robotic pitch influence a listener's autonomic responses during storytelling. By choosing a well-known story, participants would not have to work to understand new content. This minimized cognitive load and helped ensure that shifts in physiological response were due to changes in the robot's voice, not the complexity or novelty of the narrative. We used the *Pepper* robot to deliver short narrative passages under three specific vocal conditions: high, low, and default pitch. EDA was recorded to capture changes in tonic arousal, and brief surveys were used to gather subjective impressions. Together, this work introduces *Evoke* as an initial framework for evaluating emotional arousal in response to robotic voice modulation.

2 BACKGROUND

Research in human–robot interaction continues to demonstrate that different user groups may respond uniquely to the vocal qualities embedded in (SARs). Understanding these differences is critical for developing reliable protocols that evaluate how robot speech influences comfort, engagement, and emotional response. ADHD is associated with a range of attention-related challenges, and many individuals also experience sensory processing differences, including heightened sensitivity to sound (Bijlenga et al., 2017). SARs have begun to support users with cognitive or emotional needs by offering therapeutic interaction, steady co-presence, and structured guidance during activities (Khan and Anwar, 2020; Moberg et al., 2024). These systems can reinforce skill development, facilitate routines, and provide consistent companionship during tasks.

Existing work in human–robot communication highlights how vocal prosody shapes perception, emotional response, and engagement (Nass and Brave, 2005b; Eyssel et al., 2012). For individuals with ADHD, difficulties interpreting social cues or sustaining attention may heighten the importance of clear, well-modulated robot speech. EDA offers an objective and continuous measure of emotional arousal (Boucsein, 2012; Dawson et al., 2007), allowing researchers to examine how subtle variations in vocal delivery influence physiological responses such as comfort, stress, or focus.

2.1 Socially Assistive Robots and Co-Presence in ADHD Support

Socially assistive robots have been explored for ADHD support, but specific auditory adaptations remain a known challenge. Recent work indicates that SARs may help replicate aspects of this co-presence. O'Connell et al. examined the use of a Pepper robot as a study companion for students diagnosed with ADHD and found that its steady, nonjudgmental presence reduced feelings of isolation and increased perceived support during work sessions (O'Connell et al., 2024). At the same time, motor noise and other auditory cues were frequently reported as distracting, with 73% of participants identifying sound as a challenge (O'Connell et al., 2024). Interestingly, some individuals later described these sounds as anchoring or stabilizing once they adapted.

These findings align with broader theories of social facilitation, which suggest that the presence of another agent can enhance performance or engagement during familiar tasks (Zajonc, 1965; Park and

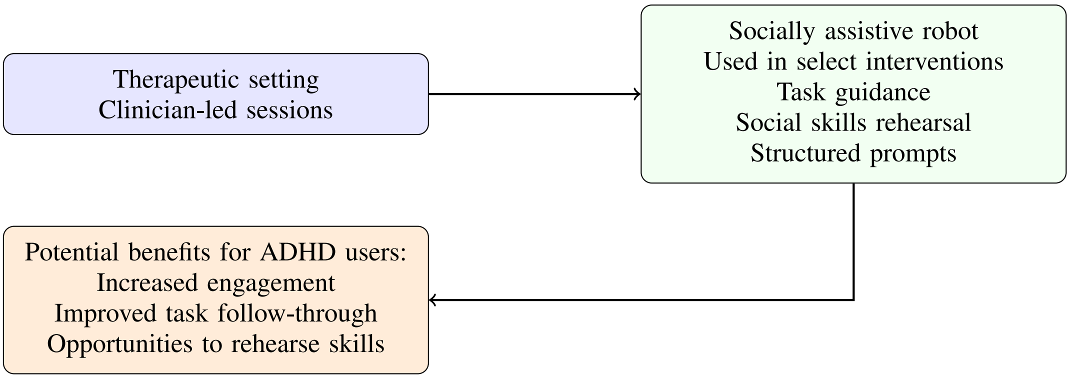

Figure 1: Conceptual view of socially assistive robots (SARs) as therapeutic tools for ADHD. In some clinician-led settings, SARs provide task structure, social rehearsal, and attention support. These systems complement traditional therapy and are not universally used.

Catrambone, 2012). Together, this research indicates that SARs may offer beneficial co-presence for individuals with ADHD, but their auditory behaviors require careful tuning to avoid overstimulation.

2.2 Voice Modulation and User Perception in HRI and the CASA Paradigm

The way a robot speaks plays a central role in shaping user perception. Aligning with the CASA paradigm, users naturally apply human-like social expectations to robots, particularly when interacting through speech (Nass and Brave, 2005b; Nass and Lee, 2001). Vocal qualities such as pitch, tone, rhythm, and pacing influence how users interpret a robot's intent, emotional state, and personality. Studies consistently find that higher-pitched or more animated voices are perceived as energetic or engaging, while lower-pitched voices convey calmness or seriousness (Cohn et al., 2020; Eyssel et al., 2012).

2.3 Research Questions

This work introduces a pilot protocol for examining how robot voice modulation influences emotional arousal. Our questions focus on feasibility and the foundations needed for future research with users who experience auditory sensitivity.

1. (RQ1) Can EDA detect changes in physiological arousal across different robot vocal conditions in a small pilot setting?
2. (RQ2) How can the insights from this pilot inform the design of future protocols for users with sensory sensitivities, including individuals with ADHD?

The motivation for this work is to better understand how robotic speech affects users who experience sensory challenges. By developing a structured approach for measuring physiological responses, this pilot provides an early framework that can support the design of socially assistive robots tailored to the needs of people with ADHD. To our knowledge, no existing protocol combines robotic voice modulation with EDA-based evaluation to inform inclusive voice design for auditory-sensitive users.

3 METHODS

This exploratory feasibility pilot examined whether robotic voice modulation could be reliably tested using a short, repeatable protocol integrating electrodermal activity (EDA) and brief self-report measures. The independent variable was robotic vocal pitch (High, Low, Default), and the dependent variable was autonomic arousal indexed by tonic skin conductance level (SCL).

3.1 Experimental Design Overview

A within-subjects design was used. Each participant listened to two or three short narrative passages adapted from the same base story. Each passage was paired with one pitch condition: High, Low, or Default. The order of presentation (High $\rightarrow$ Low $\rightarrow$ Default) was fixed to simplify the feasibility protocol. Participants could stop at any time, resulting in partial but usable within-subject coverage. After each story, participants completed a brief 3-item Likert questionnaire rating emotional valence, engagement, and perceived naturalness of the voice.

A ten-minute resting baseline preceded all storytelling, and two-minute breaks separated each condition. EDA was recorded continuously during baseline, each story, and rest periods.

3.2 Voice Modulation Conditions

Pitch manipulation used Pepper's native Choregraphe presets (*Default*, *Low Pitch*, *High Pitch*). Figure 2 illustrates the distribution of tonic skin conductance levels across baseline and the three pitch conditions. These settings were pre-tested to confirm that they produced consistent perceptual differences aligned with their intended emotional tone.

3.3 Participants

Four volunteers (n = 4) with minimal prior exposure to Pepper participated. Demographic variables were not collected in this pilot to maintain focus on protocol feasibility; future studies will incorporate broader sampling.

3.4 Story Content

Three affective variants of a short narrative (neutral, happy, and sad) were created from the same underlying plot. Each version differed only in tone and linguistic framing necessary to support the intended vocal condition. Narrative complexity was held constant to prevent confounding cognitive load with vocal pitch. Stories were approximately five minutes long.

3.5 Data Collection Procedures

Participants were fitted with an Empatica E4 wrist sensor and seated in a quiet, temperature-controlled room. After baseline recording, they listened to the story segments in the fixed order described above. EDA was captured continuously, and self-report ratings were recorded following each condition. Only high-quality data segments were retained for analysis.

3.6 Electrodermal Activity Measurement

EDA was recorded via the Empatica E4 to capture changes in autonomic arousal. Tonic SCL was the primary feature used; phasic SCRs were collected but not analyzed due to the narrative nature of the task. All EDA values were normalized to each participant's baseline to reduce inter-individual variability. Physiological sensing such as electrodermal activity (EDA) continues to be widely used in affective computing and human–computer interaction to capture implicit emotional responses during interactive tasks (Kreibig, 2023).

3.7 Self-Reported Emotional Responses

Following each story, participants rated perceived valence, engagement, and vocal naturalness on 5-point Likert scales. These subjective responses complemented the physiological measures and supported exploratory interpretation.

3.8 Experimental Controls

Environmental conditions (lighting, temperature, background noise) were held constant across sessions. Story content and vocal settings were standardized and pre-tested. Breaks were built into the protocol to reduce carryover effects.

3.9 EDA Preprocessing and Feature Extraction

EDA data were cleaned and normalized using standard procedures: removal of non-physiological values, mild median filtering, segmentation into one-minute windows, and retention of high-quality segments. Normalized tonic SCL values were averaged within each pitch condition and compared with baseline. Percent change from baseline was computed to support condition-level comparisons.

3.10 Statistical Analysis

Given the small sample size, all analyses were exploratory. Welch's t-tests compared tonic SCL in each pitch condition with baseline. Descriptive statistics and change scores supported interpretation of trends rather than inferential claims.

3.11 Ethical Considerations

All participants provided informed consent and were free to withdraw at any time. The study involved no identifiable personal data and met institutional guidelines for minimal-risk research.

4 FINDINGS

This section reports the empirical changes in tonic skin conductance level (SCL) during robot-delivered storytelling under three pitch conditions: High, Low, and Default. Analyses compare each condition to its corresponding baseline period. Only processed one-minute windows passing quality control were included.

4.1 Descriptive Tonic SCL Patterns

Mean tonic SCL values were computed for baseline and for each narration condition. All three pitch conditions exhibited lower tonic SCL than baseline. Table 1 summarizes the observed means, standard deviations, and sample sizes.

To quantify the magnitude of change relative to resting activity, difference scores were calculated as (Story − Baseline). Table 2 lists these change values. The largest decrease was observed in the Low-pitch condition, followed by Default and High pitch.

These descriptive trends indicate that the Low-pitch narration produced the largest reduction in tonic SCL, while High pitch yielded the smallest change.

4.2 Inferential Comparisons

Welch's unequal-variance t-tests were conducted to compare tonic SCL during each narration condition against baseline.

For the High-pitch condition, tonic SCL was lower than baseline but did not reach statistical significance ($t = 1.750$, $p = 0.09045$, $n_{\text{baseline}} = 27$, $n_{\text{high}} = 18$).

For the Low-pitch condition, tonic SCL was significantly lower than baseline ($t = 4.023$, $p = 0.00135$, $n_{\text{baseline}} = 27$, $n_{\text{low}} = 6$). This represents the only statistically reliable reduction among the three pitch types.

For the Default-pitch condition, a decrease in tonic SCL was observed, but the difference was not statistically significant ($t = 1.216$, $p = 0.23405$, $n_{\text{baseline}} = 27$, $n_{\text{default}} = 22$). Variability in Default-pitch windows was relatively high compared to the other conditions.

4.3 Visualization

Figure 2 illustrates the distribution of tonic SCL values across baseline and the three pitch conditions. The plot shows a consistent downward shift across narration conditions, with the Low-pitch condition showing the most pronounced reduction.

4.4 Correlation Between Physiological Arousal and Subjective Experience

To explore the relationship between physiological arousal and subjective emotional engagement, tonic SCL change scores ($\Delta =$ Story–Baseline) were compared with self-reported engagement and satisfaction ratings. Because survey responses were not originally linked to participant IDs, correlation analysis was performed only after matching survey entries to their corresponding physiological data.

Using Pearson correlation, preliminary results suggested a weak correspondence between tonic arousal changes and self-reported engagement. Participants who reported higher engagement tended to show smaller decreases in tonic SCL, whereas participants who rated engagement as low exhibited larger decreases in tonic arousal. Due to the small sample size, these findings are interpreted as exploratory rather than conclusive.

Together, the combined physiological and subjective data indicate that while robotic voice pitch influenced tonic arousal, perceived engagement and emotional effectiveness were influenced by both vocal tone and narrative content. These preliminary trends address **RQ2** by showing which vocal conditions may be calming or overstimulating, which can guide the design of future protocols for users who have auditory sensitivity or ADHD-related sensory challenges.

4.5 Overall Pattern

These outcomes indicate that tonic skin conductance levels during storytelling were lower than baseline across all three narration conditions. The extent of this reduction differed by pitch, with the largest decrease observed during Low-pitch narration, followed by Default and High pitch. Only the Low-pitch condition showed a statistically significant reduction relative to baseline, while High-pitch and Default narration also exhibited downward shifts that did not reach statistical significance based on the window-level Welch tests.

These patterns reflect both the observed mean differences and the variability and sample sizes associated with each pitch condition. Because the pilot dataset includes unequal window counts and incomplete pitch coverage across participants, the findings should be interpreted cautiously. Nonetheless, the results clearly demonstrate that tonic SCL responses were not uniform across pitch conditions, indicating measurable differences in physiological activity depending on the narration style used in the robotic delivery. These outcomes address **RQ1** by showing that EDA was sensitive to changes in vocal pitch and detected distinct differences in tonic arousal across conditions, even in a small pilot sample.

Table 1: Tonic SCL descriptive statistics by condition (in μS). Means, standard deviations, and window counts are reported for all valid segments.

Condition	Mean	Standard Deviation	Count
Baseline	0.150021	0.153213	27
Default Story	0.054957	0.339512	22
High Story	0.096550	0.034022	18
Low Story	−0.033127	0.084977	6

Table 2: Change scores (Δ = Story − Baseline) for tonic SCL. Negative values indicate reductions in tonic activity relative to baseline.

Pitch	Story Mean	Baseline Mean	Δ
Default	0.054957	0.150021	−0.095065
High	0.096550	0.150021	−0.053471
Low	−0.033127	0.150021	−0.183149

5 DISCUSSION

This *exploratory feasibility pilot* provides early insight into how variations in robotic vocal pitch influence physiological arousal during a short narrative interaction. The results showed clear differences between voice conditions, although responses varied across participants. Low-pitched vocalizations produced the most consistent calming effect, reflected in meaningful reductions in tonic skin conductance relative to baseline. This pattern suggests that lower pitch may support emotional regulation during robot-guided tasks, particularly in contexts where sustained attention and comfort are required. Because this study used a very small sample (N=4), these observations should be interpreted as directional patterns rather than definitive effects. Nonetheless, the consistent downward shifts in tonic SCL across participants indicate that the protocol is capable of detecting meaningful autonomic changes associated with voice modulation, supporting its feasibility as an evaluative framework.

Responses to the default pitch were more moderate. Although most participants showed reductions in arousal compared to baseline, the magnitude of change varied, and the effects were not statistically reliable. This variability may reflect differences in how individuals interpret neutral or unmodified robotic speech or the degree to which they are accustomed to synthetic voices more generally, highlighting the importance of baseline familiarity in future protocol design.

High-pitched vocalizations generated the greatest intersubject variability. Some participants exhibited increased arousal, while others showed decreased or minimally changed responses. These mixed outcomes indicate that high pitch may be more sensitive to personal preference, auditory sensitivity, or moment-to-moment attentional factors. The variability aligns with broader HRI findings showing that vocal expressiveness can be interpreted very differently depending on user expectations and sensory profiles. The mixed responses are particularly relevant for neurodivergent users who experience auditory sensitivity, such as individuals with ADHD. Prior work shows that heightened reactivity to sound can amplify emotional fluctuations, suggesting that high-pitch or highly animated robotic speech may require careful tuning for these populations to avoid unintended overstimulation.

Subjective feedback further supported the feasibility of the protocol. Participants reported perceiving emotional differences between conditions and expressed clear preferences for certain vocal qualities. While the small sample size limits strong inferences, the alignment between participant impressions and their qualitative physiological patterns suggests that combining self-report with EDA provides a useful basis for evaluating voice modulation in SARs. This multimodal approach strengthens confidence that observed physiological changes reflect meaningful experiential differences rather than measurement noise.

5.1 Limitations and Future Work

The most significant limitation of this pilot study is the extremely small sample size, which restricts the statistical power of the analysis and increases sensitivity to individual variability. As a result, the reported patterns should be viewed as preliminary indicators used to validate the feasibility of the protocol rather than definitive findings and should not be interpreted as conclusive evidence of causal voice modulation effects. The limited demographic diversity and short exposure times also constrain generalizability. Nonetheless, the combined use of EDA and self-reported impressions demonstrates the viability of the approach for capturing moment-to-moment emotional responses during robot-delivered narratives and

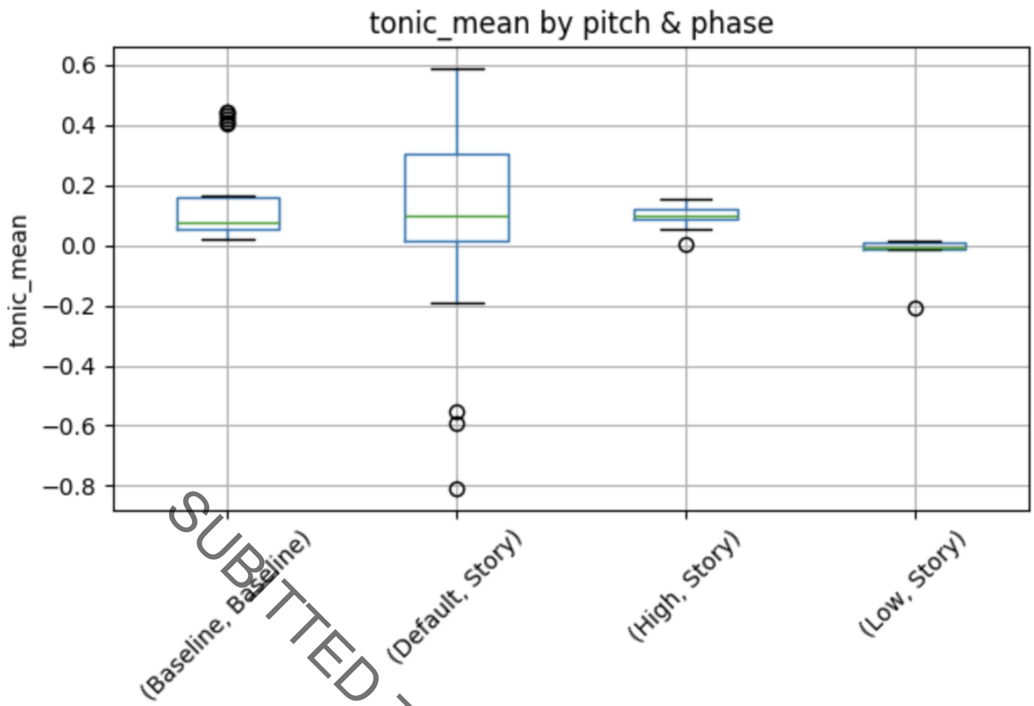

Figure 2: Distribution of tonic SCL values across baseline and pitch conditions. Negative values indicate lower conductance following preprocessing normalization.

supports its use as a foundation for larger-scale validation studies.

The findings point to a nuanced relationship between vocal pitch and emotional engagement. Low pitch appears to reliably reduce arousal, while high pitch produces heterogeneous responses that may reflect differences in auditory sensitivity or personal preference. Default pitch produced modest reductions in arousal, but additional testing is needed to clarify its role as a neutral or baseline vocal condition in SAR interaction.

Future work should recruit a larger and more diverse sample, including individuals diagnosed with ADHD who have auditory sensitivities. This would assist the HRI community in gaining a deeper understanding of how neurodivergent traits influence reactions to robotic speech. Longer and more varied narrative interactions may help capture a broader range of emotional dynamics, including habituation or fatigue effects. Expanding the protocol to incorporate additional speech parameters would be essential for maximizing comfort, focus, and regulation. Essential elements such as prosody, tempo, intensity, and timbre ought to be incorporated into future research studies to better reflect the complexity of naturalistic robotic speech.

Ultimately, these findings support the development of adaptive robotic voice systems that adjust vocal qualities in response to user state, informed by physiological feedback. Such systems hold promise for therapeutic and educational applications where emotional sensitivity, sustained engagement, and comfort are critical components of effective HRI, particularly for users with sensory processing differences.

6 CONCLUSION

This pilot study introduced Evoke, a reproducible protocol for examining how robotic vocal pitch influences tonic electrodermal activity during narrative listening. By combining controlled variations in robotic voice with objective physiological measurement, the protocol provides a structured method for assessing the autonomic impact of vocal characteristics in human–robot interaction.

Across participants, tonic SCL tended to decrease during the Low-Pitch condition compared to baseline, suggesting that lower-frequency robotic speech may promote a more relaxed autonomic state. Default-pitch narration also produced reductions in SCL, though these patterns varied more widely. High-Pitch narration showed mixed responses, indicating potential individual differences in how users interpret brighter or more energetic vocal tones. These findings demonstrate that robotic voice modulation can influence sustained physiological activity while also suggesting the need for larger and more balanced datasets to characterize these effects with greater precision.

The Evoke framework establishes a foundation for future investigations into how specific acoustic properties of robot speech shape human physiological responses. As larger studies are conducted, this methodology can support the development of adaptive vocal behaviors for socially assistive robots, particularly in contexts where users may have sensory or affective sensitivities. Continued research will be essential for refining robotic communication strategies and advancing the design of voice-based interaction in socially assistive systems.

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